INDEPENDENT TECHNICAL BRIEF

Nano Lab: Pixel 10 Pro vs Pixel 11 Pro XL

Concise matched-device comparison - saved August 19 baseline vs August 21 evaluation

The Pixel 11 Pro XL was much faster in the matched text tests, improved one real-photo object identification, and retained the same central limitation: fluent, repeatable output can still omit important information.

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Executive comparison

5.5x	0.59 s	6 / 6
Faster first fixed Prompt run	Pixel 11 proofreading mean	Capabilities worked offline on both

Area	Pixel 10 Pro	Pixel 11 Pro XL	Comparison
First fixed Prompt	5.53 s	1.00 s	Pixel 11 was 81.9% lower latency in the matched run
Top-K mean	5.72 s	0.95 s	Pixel 11 was 83.4% lower across one run at K=1, 3, 10
Summarization	1.91 s average; central results omitted	0.58 s; central trial outcome and support captured	Faster and more useful output on Pixel 11 in these captured runs
Rewriting	5.21 s average; added Thanks/[Your Name]	1.08 s; no sign-off	Faster; unwanted sign-off was absent
Proofreading	1.04 s average; four identical complete corrections	0.59 s average; four identical complete corrections	Both strong; Pixel 11 was about 43% lower latency
Synthetic image	Recognized scene; omitted tree in four identical runs	Recognized scene; omitted tree	Same completeness failure
Real photo	Charging stand called smartphone; mouse generic	Mouse identified; charging stand omitted; desk called dark	Improved mouse recognition, but still incomplete
Speech	Occasional substitutions	Connected runs meaning-preserving; two offline runs had substitutions	Both require review; live audio prevents precise device ranking
Offline	All six worked after restart	All six worked after restart	Same practical on-device result after assets were installed

INTERPRET The Pixel 11 Pro XL showed a clear latency advantage in these exact captured tests. This is a device-specific result from one unit per model, not a population benchmark or a claim about every Pixel.

1. Test basis and comparability

Item	Pixel 10 Pro baseline	Pixel 11 Pro XL session
Date	August 19, 2026	August 21, 2026
Codename	blazer	kodiak
OS / SDK	Android 17 / SDK 37	Android 17 / SDK 37
Build	google/blazer/blazer:17/CP2A.260805.005/15828068:user/release-keys	google/kodiak/kodiak:17/CD1A.260714.001.A9/15938155:user/release-keys
RAM	16 GB advertised; about 15.21 GiB accessible	16 GB advertised; accessible total not separately measured
App	Release build	Transferred release APK; versionName 1.0.0; code 1; targetSdk 36
AICore	Main and isolated-service processes observed	Main and isolated-service processes observed
Power	Unplugged; 100%; 37.1 C; thermal None	Intended unplugged; 100%; 31.3 C; thermal None
Observation method	Read-only ADB	Read-only ADB

The Pixel 10 Pro was being factory-reset and traded in, so saved results are the authoritative baseline and could not be rerun. The Pixel 11 Pro XL inherited an APK from the Pixel 10 Downloads transfer; the exact repository commit was not recorded.

Matched latency observations

Test	Pixel 10	Pixel 11	Observed reduction / speedup
First fixed Prompt	5.53 s	1.00 s	81.9% lower / 5.53x
Top-K 1	5.86 s	0.87 s	85.2% lower / 6.74x
Top-K 3	5.80 s	1.07 s	81.6% lower / 5.42x
Top-K 10	5.51 s	0.91 s	83.5% lower / 6.05x
Three candidates	9.70 s	1.65 s	83.0% lower / 5.88x
Dedicated summary	1.91 s average	0.58 s	69.6% lower / 3.29x
Professional rewrite	5.21 s average	1.08 s	79.3% lower / 4.82x
Proofreading	1.04 s average	0.59 s average	43.3% lower / 1.76x
Synthetic image	4.04 s average	1.37 s	66.1% lower / 2.95x
Real photograph	3.75 s average	1.27 s	66.1% lower / 2.95x

LIMIT

Several Pixel 10 values are multi-run averages while some Pixel 11 values are single runs. Top-K has one run per value on each device. These arithmetic comparisons describe the captured records and are not confidence intervals or statistically controlled performance claims.

2. Prompt behavior

Both devices followed the uppercase and exact-three-sentence instructions in the controlled Top-K comparison. On both, changing only Top-K changed wording, but a higher setting was not necessarily better. One run per value prevents a repeatability or monotonic-trend conclusion.

Control	Pixel 10 Pro	Pixel 11 Pro XL
Temperature	Temperature 0 was stable; higher temperature varied wording	Temperature changed wording across the captured runs
Fixed seed	Identical outputs	Two identical outputs at temp 1.0, seed 123
Seed 0	Variation observed	Two different outputs at temp 1.0
Multiple candidates	Distinct candidates	Three distinct candidates
Structured JSON	All facts extracted, but unwanted Markdown fences added	Not rerun in the completed Pixel 11 dataset
ISO date sorting	Three identical temperature-0 runs; all facts preserved	Not rerun because exact prior input/settings were not retained

LIMIT The date-sorting result cannot be treated as a matched Pixel 11 comparison. The old Pixel 10 Pro could not be rerun, and no replacement input was invented.

3. Quality differences by capability

Summarization

The Pixel 10 dedicated summary was fast and factual but consistently omitted the article's central results. The Pixel 11 output - 'Alder Cove launched a successful tool library trial; residents strongly support its continuation for permanent funding.' - directly captured the trial outcome and resident support. This is an observed output improvement on the fixed article, not evidence that every Pixel 11 summary will prioritize better.

Rewriting

The Pixel 10 highest-confidence rewrite preserved facts but consistently added 'Thanks' and '[Your Name].' The Pixel 11 highest-confidence rewrite preserved the same factual content and did not add a closing. It returned three suggestions in 1.08 seconds.

Proofreading

Both devices produced four identical, fact-preserving corrections. Pixel 11 averaged 0.59 seconds versus about 1.04 seconds on Pixel 10. However, the visible Pixel 11 input used 'labeled' while the Pixel 10 input used the misspelling 'labeld,' so the inputs were not character-for-character identical.

4. Image Description and Speech Recognition

Synthetic image

Device	Recorded description behavior
Pixel 10 Pro	Accurately described the red house scene but omitted the prominent tree in four identical runs.
Pixel 11 Pro XL	Accurately described the house, colors, clouds, and sun but again omitted the tree; also called the flat green ground a hill.

INTERPRET

The same salient omission across both devices suggests a limitation of the tested concise Image Description behavior or model path on this fixed scene. It does not prove every Nano version or Pixel will omit the object.

Real tabletop photograph

Object/detail	Pixel 10 Pro	Pixel 11 Pro XL
Computer mouse	Reduced to 'a small black object'	Explicitly identified as a black mouse
Charging stand	Confidently misidentified as a smartphone	Omitted completely
Desk/tabletop	Correctly called gray	Incorrectly generalized as dark-colored
Repeatability	Same output in four runs	Same output in the initial plus three additional online runs

A separate one-off Gemini 3.7 Flash web-app check recognized all objects in the photograph. That cloud result is qualitative context only; it was not Nano Lab execution, timing, or offline evidence.

Advanced Speech Recognition

Pixel 10 produced occasional meaningful substitutions, including 'Three' to 'they,' 'labeled' to 'labor,' and 'fictional' to 'fixed.' Pixel 11's two connected runs preserved meaning at slow and normal pace. Its two offline runs inserted 'at' before 2026 in one run and changed 'received' to 'receives' in another.

LIMIT

Speech used live microphone input, so exact acoustic conditions and spoken delivery were not identical. Session time includes speaking and manual interaction. The comparison supports a user-review requirement, not a formal word-error-rate ranking.

5. Memory, downloads, and offline operation

AICore memory snapshots

Checkpoint	Pixel 10 AICore PSS / RSS	Pixel 11 AICore PSS / RSS
1. Before app	208.7 / 435.8 MiB	118.8 / 351.0 MiB
2. App open; clients ready	201.5 / 431.9 MiB	119.1 / 353.9 MiB
3. After Stable availability	201.7 / 432.4 MiB	Not captured
4. After first fixed Prompt	298.9 / 531.9 MiB	146.1 / 383.0 MiB

Nano Lab app PSS

Checkpoint	Pixel 10	Pixel 11
2. App open; clients ready	192.2 MiB	155.3 MiB PSS; 283.7 MiB RSS; 17.2 MiB swap PSS
3. After Stable availability	241.6 MiB	Not captured
4. After first fixed Prompt	253.1 MiB	108.8 MiB PSS; 223.5 MiB RSS; 39.4 MiB swap PSS

INTERPRET

Pixel 11 had lower observed AICore PSS and RSS at the three comparable captured points. Because these are Android snapshots with different process state, cache, swap, graphics, and background conditions, they do not establish total model size, reserved RAM, or a causal hardware advantage.

Download behavior

- Pixel 10 initially reported the Prompt model downloadable, then available after download/restart; Advanced Speech downloaded about 97.3 MiB.
- Pixel 11's Prompt model became available after an initial request, but Nano Lab left a stale progress panel visible until relaunch.
- Pixel 11's Advanced Speech transfer was 22.4 MiB and displayed correct progress and completion.
- The different visible download sizes may reflect device/build/model asset state. They are observations, not evidence that the complete model footprint differs by the same amount.

Offline result

After required assets were installed and each phone was restarted, all six capabilities remained available and completed representative tests with airplane mode on and Wi-Fi off. This is strong practical evidence of on-device operation. It is not formal network forensics, and AICore may use connectivity for asset management when connectivity is available.

6. Conclusions and disclosure

Directly supported conclusions

- The Pixel 11 Pro XL completed the matched text tasks much faster than the saved Pixel 10 Pro baseline in this controlled dataset.
- Both devices showed strong deterministic behavior under fixed conditions and supported all six tested capabilities offline after assets were installed.
- Pixel 11 improved real-photo mouse recognition and avoided Pixel 10's false smartphone assertion, but still omitted the charging stand and introduced a desk-color error.
- Both devices omitted the prominent tree from the synthetic scene.
- Neither device's speech output should be treated as exact without review.
- Repeatability must not be confused with correctness or completeness.

Limitations

- One physical unit of each model was tested; findings are device-specific and should not be generalized to every Pixel unit.
- The Pixel 10 Pro was unavailable for reruns because it was being factory-reset and traded in; its saved report is authoritative.
- The small test set was designed to expose practical behavior, not produce statistical benchmark claims.
- Pixel 11 memory checkpoint 3 was missed after the Prompt download UI defect and relaunch.
- Proofreading inputs differed by one misspelling, and date sorting was not rerun on Pixel 11 because the exact prior input/settings were not retained.
- Model, AICore, Android, and beta/alpha API updates can change future behavior.

> The best reading of the comparison is not that the newer phone made Nano infallible. It made the tested assistance markedly faster while leaving the need for human validation intact.

Development and AI-assistance disclosure

Jason McArdle conceived and directed Nano Lab, defined the evaluation questions, made all project and testing decisions, performed every physical-device test, supplied the inputs, reviewed the outputs, identified problems, directed revisions, and takes responsibility for the findings and publication.

ChatGPT served as a collaborative tool for implementation support, API research, troubleshooting, data organization, interpretation, and drafting. McArdle reviewed and directed that work and made the final publishing decisions. All measurements and outputs came from Nano Lab on physical Pixel hardware; none were simulated or AI-generated.